King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 102 Major Exam I Term 253

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. $\int_1^4 \dfrac{1}{\sqrt{x}\,(1+\sqrt{x})}\,dx =$

(a) $2\ln\dfrac{3}{2}$

(b) $2\ln\dfrac{2}{3}$

(c) $\ln\dfrac{2}{3}$

(d) $\ln\dfrac{3}{2}$

(e) $\ln\dfrac{4}{3}$

2. $\lim_{n\to\infty}\sum_{i=1}^{n}\left(\dfrac{i^4}{n^5}+\dfrac{i}{n^2}\right) =$ (Hint: Recognize the sum as a Riemann sum for a function defined on $[0,1]$.)

(a) $\dfrac{3}{5}$

(b) $\dfrac{1}{5}$

(c) $\dfrac{9}{10}$

(d) $\dfrac{7}{10}$

(e) $\dfrac{4}{5}$

3. The area of the region bounded by the graphs of $f(x)=\sin x$ and $g(x)=\dfrac{2}{\pi}x$ on the interval $[0,\pi]$ is equal to

(a) 2π

(b) $\dfrac{\pi}{3}$

(c) $\dfrac{\pi}{4}$

(d) $\dfrac{\pi}{2}$

(e) π

4. The average value of $f(x) = 3x^2 + 8x$ on the interval $[-1, 2]$ is equal to

(a) 7

(b) 6

(c) 3

(d) 4

(e) 5

5. $\int_0^{1/6} \dfrac{3}{\sqrt{1-9x^2}}\, dx =$

(a) $\dfrac{\pi}{6}$

(b) $\dfrac{\pi}{18}$

(c) $\dfrac{\pi}{9}$

(d) $\dfrac{\pi}{2}$

(e) $\dfrac{\pi}{3}$

6. If $F(x) = \int_0^{\tan x} \sqrt{t + \sqrt{t}}\, dt$, then $F'\left(\dfrac{\pi}{4}\right) =$

(a) 1

(b) $4\sqrt{2}$

(c) $\dfrac{1}{\sqrt{2}}$

(d) $\sqrt{2}$

(e) $2\sqrt{2}$

7. $\int_{-\pi/2}^{\pi/2} \sin(7x)\cos(3x)\, dx =$

(a) 1

(b) $\dfrac{\pi}{4}$

(c) π

(d) $\dfrac{\pi}{2}$

(e) 0

8. Using the Midpoint rule with 2 rectangles, the approximate area of the region bounded by the graph of $f(x) = \cos x$ and the x-axis over the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is equal to

(a) $\frac{\pi}{2}$

(b) π

(c) $\sqrt{2}\,\pi$

(d) $\frac{\pi}{\sqrt{2}}$

(e) 2π

9. $\int \frac{x^2+1}{x+1}\,dx =$

(a) $\frac{x^2}{4} - x + 2\ln|x+1| + C$

(b) $\frac{x^2}{2} - x + 2\ln|x+1| + C$

(c) $\frac{x^2}{2} - x - 2\ln|x+1| + C$

(d) $\frac{x^2}{2} + x - 2\ln|x+1| + C$

(e) $\frac{x^2}{2} + x + 2\ln|x+1| + C$

10. Consider an odd continuous function f such that $\int_{-3}^{6} f(x)\,dx = 9$ and $\int_{0}^{3} f(x)\,dx = -3$. Then $\int_{0}^{6} f(x)\,dx =$

(a) -12

(b) 0

(c) 12

(d) -6

(e) 6

11. $\int_{0}^{\ln 3} \tanh x\,dx =$

(a) $\ln\frac{5}{3}$

(b) $\ln\frac{4}{3}$

(c) $\ln\frac{3}{5}$

(d) $\ln\frac{3}{4}$

(e) $\ln\frac{5}{4}$

12. The area of the region bounded by the graphs of the functions $f(x) = x(2-x)$ and $g(x) = -x$ is equal to

(a) 4

(b) $\frac{7}{2}$

(c) 3

(d) $\frac{5}{2}$

(e) $\frac{9}{2}$

13. $\int_0^1 x(1-x)^9\,dx =$

(a) $\frac{1}{101}$

(b) $\frac{1}{9}$

(c) $\frac{1}{10}$

(d) $\frac{1}{110}$

(e) $\frac{1}{100}$

14. The lower sum (in terms of n, the number of subintervals) for the region bounded by the graph of $f(x) = 4 - x^2$ and the x-axis on the interval $[0, 2]$ is

(a) $\frac{3}{4n^2}\left(4n^2 + 3n - 1\right)$

(b) $\frac{4}{3n^2}\left(4n^2 - 3n - 1\right)$

(c) $\frac{2}{n^2}\left(4n^2 - 3n - 1\right)$

(d) $\frac{3}{4n^2}\left(4n^2 - 3n + 1\right)$

(e) $\frac{4}{3n^2}\left(4n^2 + 3n + 1\right)$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(a)**	$u = 1 + \sqrt{x}$: $\int_2^3 \frac{2}{u}\,du = 2\ln\frac{3}{2}$.
2	**(d)**	Riemann sum for $\int_0^1 (x^4 + x)\,dx = \frac{1}{5} + \frac{1}{2} = \frac{7}{10}$.
3	**(d)**	Cross at $0, \frac{\pi}{2}$: $\left(1 - \frac{\pi}{4}\right) + \left(\frac{3\pi}{4} - 1\right) = \frac{\pi}{2}$.
4	**(a)**	$\frac{1}{3}\int_{-1}^2 (3x^2 + 8x)\,dx = \frac{1}{3} \cdot 21 = 7$.
5	**(a)**	$u = 3x$: $\arcsin u\big\|_0^{1/2} = \frac{\pi}{6}$.
6	**(e)**	$F'(x) = \sqrt{\tan x + \sqrt{\tan x}}\sec^2 x$; at $\frac{\pi}{4}$: $\sqrt{2} \cdot 2 = 2\sqrt{2}$.
7	**(e)**	Integrand $= \frac{1}{2}(\sin 4x + \sin 10x)$ is odd on a symmetric interval, so the integral is 0.
8	**(d)**	$\Delta x = \frac{\pi}{2}$, midpoints $\pm\frac{\pi}{4}$: $2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\pi}{2} = \frac{\pi}{\sqrt{2}}$.
9	**(b)**	$\frac{x^2+1}{x+1} = x - 1 + \frac{2}{x+1}$, so $\frac{x^2}{2} - x + 2\ln\lvert x+1\rvert + C$.
10	**(e)**	Odd: $\int_{-3}^3 f = 0$, so $\int_3^6 f = 9$ and $\int_0^6 f = -3 + 9 = 6$.
11	**(a)**	$\ln\cosh x\big\|_0^{\ln 3} = \ln\frac{5}{3}$, since $\cosh(\ln 3) = \frac{5}{3}$.
12	**(e)**	Intersections $x = 0, 3$: $\int_0^3 (3x - x^2)\,dx = \frac{9}{2}$.
13	**(d)**	$u = 1 - x$: $\int_0^1 (u^9 - u^{10})\,du = \frac{1}{10} - \frac{1}{11} = \frac{1}{110}$.
14	**(b)**	f decreasing: right endpoints, $s_n = 8 - \frac{4(n+1)(2n+1)}{3n^2} = \frac{4}{3n^2}(4n^2 - 3n - 1)$.